

Artificial Intelligence and Machine Learning

Introduction

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence. Machine Learning (ML) is a subset of AI that enables computers to learn from data without being explicitly programmed.

Types of Machine Learning

1. Supervised Learning: Uses labeled data. Examples: Spam Detection, House Price Prediction.
2. Unsupervised Learning: Uses unlabeled data. Examples: Customer Segmentation, Clustering.
3. Reinforcement Learning: Learns using rewards and penalties. Applications include robotics and self-driving cars.

Deep Learning

Uses neural networks such as ANN, CNN, RNN, LSTM, and Transformers for image recognition, speech recognition, NLP, and medical diagnosis.

Natural Language Processing (NLP)

Applications include chatbots, language translation, summarization, sentiment analysis, and question answering.

Retrieval-Augmented Generation (RAG)

RAG combines document retrieval with Large Language Models to answer questions accurately using external documents.

Main Components

- Document Loader
- Text Splitter
- Embedding Model
- Vector Database (FAISS, ChromaDB, Pinecone)
- Retriever
- Large Language Model (Llama 3, GPT-4, Gemini)

Advantages

Higher accuracy, reduced hallucination, works with private documents, and fast retrieval.

Applications

Research, education, customer support, enterprise knowledge bases, legal document analysis, and healthcare.

Sample Questions

1. What is AI?
2. What is Machine Learning?
3. Explain RAG.
4. What are the components of a RAG system?
5. Name three vector databases.